

CE 403: Wastewater Engineering

Course Code	CE 403
Course Name	Wastewater Engineering
L-T-P-C	3-0-0-3
Prerequisites	IC 230
Approval	36th BoA (Sep 2020); 28th Senate (Nov 2020), in the B.Tech Civil Engineering curriculum

Course Contents

- **Planning and hydraulic design of sewerage system**
Population forecasting of town, equivalent sewage flow estimation, onsite, decentralized and centralized systems, Sewer pipelines and the materials, sewers hydraulics, Storm drainage, Storm runoff estimation, sewer appurtenances, corrosion in sewers its prevention and control, sewage pumping, drainage in buildings, plumbing systems for drainage, wastewater Collection System, Instrumentation and automation techniques. [8 Lectures]
- **Primary treatment Processes**
Unit Operations and Processes, Selection of treatment processes, Onsite sanitation, Septic tank, Grey water harvesting, Primary treatment, Principles, functions and design of sewage treatment units, screens, grit chamber, primary sedimentation tanks, Construction, Operation and Maintenance aspects. [8 Lectures]
- **Secondary treatment Processes**
Selection of Treatment Methods, Hydraulic Principles, Functions Activated Sludge Process and Extended aeration systems, Trickling filters and their design, Sequencing Batch Reactor (SBR), Membrane Bioreactor (MBR), Waste Stabilization Ponds, Reclamation and Reuse of sewage, Recent Advances in Sewage Treatment, Construction, Operation and Maintenance aspects. [10 Lectures]
- **Sewage Disposal, Recycle and Reuse**
Standards for Disposal Methods, dilution, Mass balance principle, Selfpurification of river, Oxygen sag curve, deoxygenation and reaeration, Dissolved oxygen Modelling, Land disposal, Agriculture uses of treated Sewage, Cycle of emerging organic contaminant and their effects on human health, sodium hazards, Soil dispersion system. [10 Lectures]
- **Sludge and Solid Waste Management**
Sludge characterization, Thickening, Design of gravity thickener, Sludge digestion, Standard rate and High rate digester design, Biogas recovery, Sludge Conditioning and Dewatering, Sludge drying beds, ultimate residue disposal. [6 Lectures]

Text Books

1. Metcalf, L., Eddy, H. P., and Tchobanoglous, G., **Wastewater engineering: treatment, disposal, and reuse (Vol. 4)**, McGraw-Hill, 1979.
2. Karia, G. L., and Christian, R. A., **Wastewater treatment: Concepts and design approach**, PHI Learning Pvt. Ltd., 2013.
3. Duggal K.N., **Elements of Environmental Engineering**, S. Chand and Co. Ltd., 2014.
4. Mackenzie L. Davis, **Water and Wastewater Engineering: Design Principles and Practice**, McGraw-Hill Education, 2010.

References

1. Metcalf, Leonard, Harrison P. Eddy, and Georg Tchobanoglous, **Waste water engineering: treatment, disposal, and reuse**, McGraw-Hill, 1979.
2. Garg, S.K, **Environmental Engineering Vol. II**, Publishers, 2015.
3. **Journal of Water Processing Engineering-Elsevier**
4. **Journal of Environmental Engineering-ASCE**